GEOSPATIAL DATA ANALYSIS

Regional Market Opportunity & Expansion Analysis

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1. Project Overview

This project analyzes synthetic city-level geographic market data across India to identify promising regional expansion opportunities using search demand, competition, lead conversion, income and estimated acquisition cost.

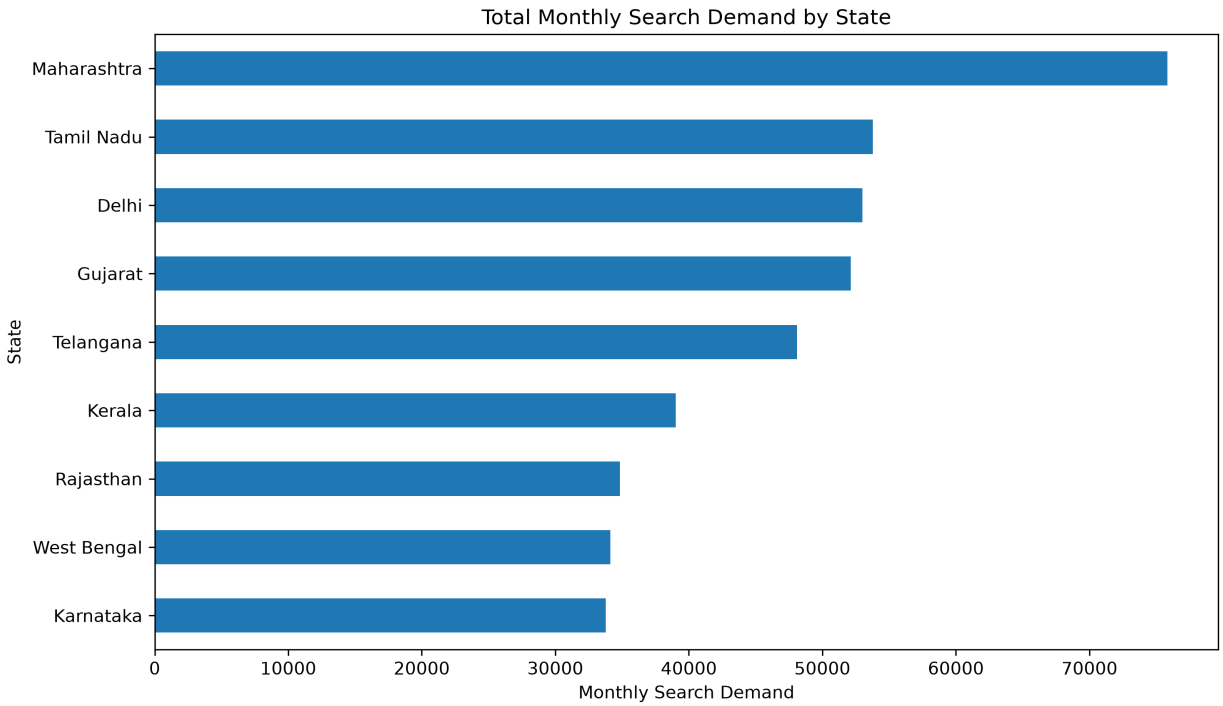
2. Dataset & Quality Validation

Metric	Result
Dataset records	400
Columns	11
Missing values	0
Duplicate rows	0
Unique Region IDs	400
Unique cities	10
Unique states	9

3. Methodology

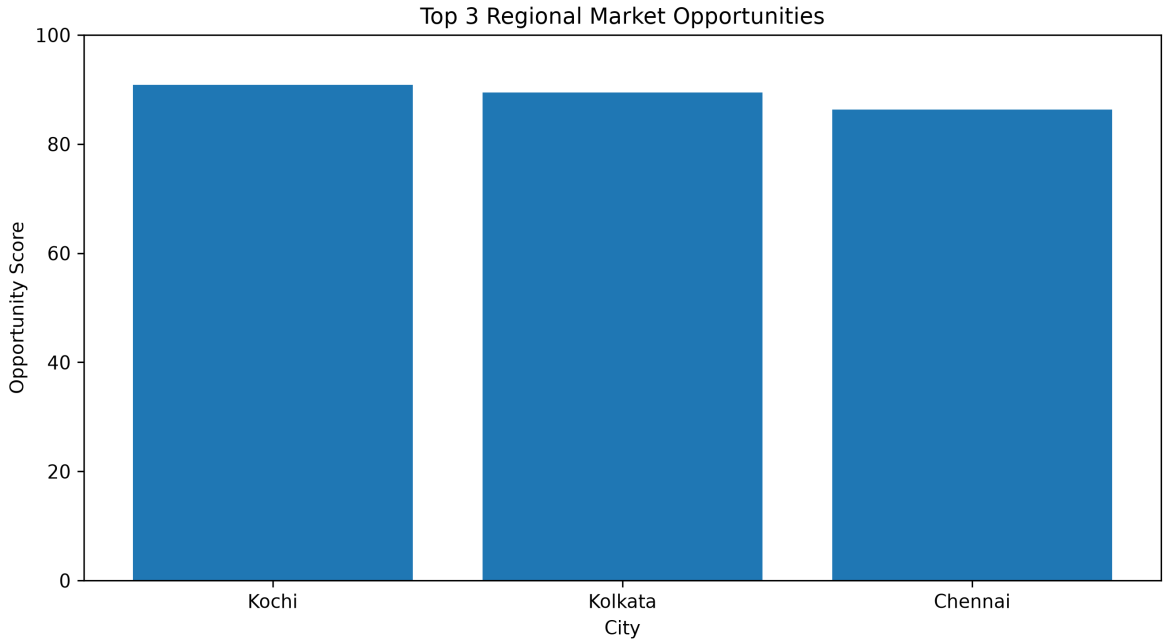
The dataset was checked for missing values, duplicates, valid coordinates and invalid numerical values. Text fields were standardized. State-level metrics were aggregated. A weighted opportunity score was calculated using normalized demand (40%), lower competition (25%), conversion rate (20%), income (10%) and lower acquisition cost (5%).

4. State-wise Market Demand



5. Top 3 Opportunity Regions

Region ID	City	State	Demand	Score
REG-0206	Kochi	Kerala	1787	90.88
REG-0353	Kolkata	West Bengal	1799	89.47
REG-0373	Chennai	Tamil Nadu	1831	86.30



6. Key Findings

• Kochi, Kerala achieved the highest opportunity score of 90.88.

• Kolkata, West Bengal ranked second with a score of 89.47.

• Chennai, Tamil Nadu ranked third with a score of 86.30.

• The selected regions combine strong search demand with relatively low competition.

7. Business Recommendations

Prioritize further market research in Kochi, Kolkata and Chennai. Validate customer demand through local surveys and pilot campaigns. Compare actual acquisition costs and conversion performance before committing expansion budgets.

8. Dataset Limitations

The dataset contains synthetic coordinates and market indicators. It does not provide actual sales revenue, postal codes, population counts or verified user density. Monthly search demand is used as a demand indicator, not as actual customer count. Region record counts indicate dataset coverage, not population density. The opportunity score is a custom weighted ranking and should not be treated as a guaranteed business outcome.

9. Project Deliverables

• Cleaned geographic dataset

• Interactive India opportunity map (HTML)

• State-level market analysis (CSV)

• Top 3 opportunity regions (CSV)

- Regional opportunity scores (CSV)
- Market demand and opportunity charts